

MINIMAL-GOOD-LATTICE NORMALIZATION OF LATTICE-MODEL SPLITTINGS

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ABSTRACT. We compare the splittings of the universal metaplectic extension furnished by generalized lattice models. Their ratios are continuous $\mathbb{C}^\times$ -valued characters satisfying a cocycle identity. A minimal good lattice on the opposite member of the dual pair then defines a choice-independent splitting, and comparison with it gives the normalized character $\xi_{x,x'}$.

1. SETUP AND KNOWN INPUTS

Let

$$(G, G') = (U(V, h), U(V', h')) \subseteq \mathrm{Sp}(W)$$

be the type-I reductive dual pair under consideration, and work in the universal extension

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \widetilde{\mathrm{Sp}}(W) \longrightarrow \mathrm{Sp}(W) \longrightarrow 1.$$

Fix $x \in \mathcal{B}(G)$ and put $K_x := \mathrm{Stab}_G(x)$. We assume that the building has been refined so that K_x is constant on each open facet. For every $x' \in \mathcal{B}(G')$, the generalized lattice model gives a continuous group-homomorphic splitting

$$s_{x,x'}: K_x \longrightarrow \widetilde{\mathrm{Sp}}(W).$$

We use the following known inputs. The nonempty set $\mathcal{M}(V')$ of minimal good lattices is a single G' -orbit. Each $\Lambda' \in \mathcal{M}(V')$ determines a normalized building point $x'_{\Lambda'}$, equivariantly in the sense that

$$x'_{a\Lambda'} = ax'_{\Lambda'} \quad (a \in G').$$

Generalized-lattice-model naturality says that, for a lift $\tilde{a}$ of $1 \otimes a$ and $g \in K_x$,

$$(1) \quad s_{x,ax'}(g) = \tilde{a} s_{x,x'}(g) \tilde{a}^{-1}.$$

Finally, the known commuting-cover theorem for this reductive dual pair says that the pullback covering groups over G and G' commute elementwise in $\widetilde{\mathrm{Sp}}(W)$. This nontrivial cover-level theorem is used as an input; it is not deduced from commutativity in $\mathrm{Sp}(W)$.

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2. COMPARISON CHARACTERS

For $x'_1, x'_2 \in \mathcal{B}(G')$ define

$$(2) \quad \xi_{x; x'_1, x'_2}(g) := s_{x, x'_1}(g) (s_{x, x'_2}(g))^{-1}, \quad g \in K_x.$$

The two factors have the same image g in $\mathrm{Sp}(W)$, so their quotient lies in the central kernel $\mathbb{C}^\times$.

Proposition 2.1. *The map $\xi_{x; x'_1, x'_2} : K_x \rightarrow \mathbb{C}^\times$ is a continuous character. For all $x'_1, x'_2, x'_3 \in \mathcal{B}(G')$ it satisfies*

$$\xi_{x; x'_1, x'_3} = \xi_{x; x'_1, x'_2} \xi_{x; x'_2, x'_3}.$$

Proof. Write $s_i = s_{x, x'_i}$ and $\xi_{ij} = \xi_{x; x'_i, x'_j}$. Since $s_i(g) = \xi_{ij}(g) s_j(g)$ and $\xi_{ij}(g)$ is central, multiplicativity of the splittings gives

$$\begin{aligned} \xi_{ij}(gh) s_j(g) s_j(h) &= s_i(gh) = s_i(g) s_i(h) \\ &= \xi_{ij}(g) \xi_{ij}(h) s_j(g) s_j(h). \end{aligned}$$

Cancellation proves $\xi_{ij}(gh) = \xi_{ij}(g) \xi_{ij}(h)$; continuity follows from continuity of the two splittings. Direct cancellation in the central kernel gives

$$s_1(g) s_3(g)^{-1} = (s_1(g) s_2(g)^{-1}) (s_2(g) s_3(g)^{-1}),$$

which is the cocycle identity. $\square$

3. THE MINIMAL SPLITTING

For $\Lambda' \in \mathcal{M}(V')$ put

$$s_{x, \Lambda'}^{\min} := s_{x, x'_{\Lambda'}}.$$

Theorem 3.1 (Independence of the minimal good lattice). *For all $\Lambda'_0, \Lambda'_1 \in \mathcal{M}(V')$,*

$$s_{x, \Lambda'_0}^{\min} = s_{x, \Lambda'_1}^{\min}$$

as continuous group-homomorphic splittings. Their common value therefore defines a canonical splitting

$$s_x^{\min} : K_x \longrightarrow \widetilde{\mathrm{Sp}}(W).$$

Proof. Choose $a \in G'$ with $\Lambda'_1 = a\Lambda'_0$, and put $x'_0 = x'_{\Lambda'_0}$. Equivariance gives $x'_{\Lambda'_1} = ax'_0$. If $\tilde{a}$ is any lift of $1 \otimes a$, then lattice-model naturality (1) gives

$$(3) \quad s_{x, ax'_0}(g) = \tilde{a} s_{x, x'_0}(g) \tilde{a}^{-1}.$$

This is only a conjugacy statement. The element $s_{x, x'_0}(g)$ lies in the pullback cover over G , while $\tilde{a}$ lies in the pullback cover over G' . The separate commuting-cover theorem therefore gives

$$[\tilde{a}, s_{x, x'_0}(g)] = 1.$$

Substitution in (3) yields $s_{x, ax'_0}(g) = s_{x, x'_0}(g)$ for every $g \in K_x$. Since $x'_{\Lambda'_1} = ax'_0$, the two minimal splittings agree. Their common value remains

continuous, group-homomorphic, and a splitting because every $s_{x,x'}$ has these properties by construction. $\square$

4. NORMALIZED CORRECTION CHARACTER

Corollary 4.1. *For arbitrary $x' \in \mathcal{B}(G')$, the formula*

$$(4) \quad \boxed{\xi_{x,x'}(g) := s_x^{\min}(g)(s_{x,x'}(g))^{-1}, \quad g \in K_x,}$$

defines a continuous $\mathbb{C}^\times$ -valued character, and

$$(5) \quad \boxed{s_x^{\min} = \xi_{x,x'} s_{x,x'}.$$

Moreover,

$$(6) \quad \xi_{x;x'_1,x'_2} = \xi_{x,x'_1}^{-1} \xi_{x,x'_2},$$

and $\xi_{x,x'_\Lambda} = 1$ for every minimal good lattice Λ' .

Proof. Equation (4) is the comparison of the two splittings $s_x^{\min}$ and $s_{x,x'}$, so the proposition proves that it is a continuous character. Rearranging its definition gives (5). Also, centrality gives

$$\begin{aligned} \xi_{x,x'_1}^{-1}(g) \xi_{x,x'_2}(g) &= s_{x,x'_1}(g)(s_x^{\min}(g))^{-1} s_x^{\min}(g)(s_{x,x'_2}(g))^{-1} \\ &= s_{x,x'_1}(g)(s_{x,x'_2}(g))^{-1} = \xi_{x;x'_1,x'_2}(g), \end{aligned}$$

which proves (6). If $x' = x'_\Lambda$, then independence of Λ' gives $s_{x,x'} = s_x^{\min}$, hence $\xi_{x,x'} = 1$. $\square$

Remark 4.2 (Scope of the normalization). The formal comparison above does not prove that $\xi_{x,x'}$ is trivial on K_x^+ , factors through K_x/K_x^+ , is quadratic, or equals a determinant product on a finite graded quotient. Each such refinement requires a separate argument.